📘 **AI/ML Internship – Day 19 Notes & Tasks**

🟡 **Module 4: NumPy (Operations – Part 2)**

📌 **Part 1: NumPy Arithmetic Operations**

👉 NumPy allows **direct operations on arrays** (no loops needed)

🔹 **Example**

```
import numpy as np

a = np.array([1, 2, 3])
b = np.array([4, 5, 6])

print(a + b)   # Addition
print(a - b)   # Subtraction
print(a * b)   # Multiplication
print(a / b)   # Division
```

🔹 **Output**

👉 Element-wise operations:

- [1+4, 2+5, 3+6] → [5, 7, 9]

📌 **Part 2: Scalar Operations**

👉 Operation with a single value

```
a = np.array([1, 2, 3])

print(a + 10)
print(a * 2)
```

📌 **Part 3: Vectorization (VERY IMPORTANT)**

🔹 **What is Vectorization?**

👉 Performing operations on entire arrays without loops

🔹 Example

```
# Without NumPy (slow)
nums = [1, 2, 3]
result = []

for x in nums:
    result.append(x * 2)

# With NumPy (fast)
arr = np.array([1, 2, 3])
print(arr * 2)
```

🔹 Why Important?

- Faster execution
- Cleaner code
- Used in ML models

📌 Part 4: Mathematical Functions

🔹 Basic Functions

```
arr = np.array([1, 4, 9])

print(np.sqrt(arr))   # Square root
print(np.exp(arr))    # Exponential
print(np.log(arr))    # Logarithm
```

🔹 Rounding

```
arr = np.array([1.2, 2.7, 3.5])

print(np.round(arr))
print(np.ceil(arr))
print(np.floor(arr))
```

📌 **Part 5: Aggregation Functions**

👉 Used for data analysis

```
arr = np.array([10, 20, 30, 40])

print(np.sum(arr))
print(np.mean(arr))
print(np.max(arr))
print(np.min(arr))
```

📌 **Part 6: Axis Concept (Important)**

👉 Used in multi-dimensional arrays

```
arr = np.array([[1, 2], [3, 4]])

print(np.sum(arr, axis=0))  # Column-wise
print(np.sum(arr, axis=1))  # Row-wise
```

📌 **Part 7: Comparison Operations**

```
arr = np.array([10, 20, 30])

print(arr > 15)
```

🔹 **Filtering**

```
print(arr[arr > 15])
```

📌 **Part 8: Real-World Thinking**

👉 In AI/ML:

- Operations → feature calculation
- Aggregation → statistics
- Filtering → data cleaning

👉 Vectorization = **speed of ML models**

🎯 **Tasks for Day 19**

🧠 **Task 1: Theory**

1. What is vectorization?
2. Difference between list and NumPy operations
3. What are aggregation functions?
4. What is axis?
5. Why NumPy is faster?

💻 **Task 2: Arithmetic Practice**

1. Add two arrays
2. Multiply arrays
3. Divide arrays

💻 **Task 3: Scalar Operations**

1. Add constant to array
2. Multiply array
3. Subtract constant

💻 **Task 4: Math Functions**

1. Find square root
2. Find log values
3. Round values

💻 **Task 5: Aggregation**

1. Find sum
2. Find mean
3. Find max and min

💻 **Task 6: Axis Practice**

1. Row-wise sum
2. Column-wise sum

💻 **Task 7: Filtering**

1. Filter values > 20
2. Filter even numbers